

APPLICATION OF LANDSCAPE PATTERN ANALYSIS TO QUANTITATIVELY EVALUATE THE SPATIAL STRUCTURE CHARACTERISTICS OF PREFERENTIAL FLOW PATHS IN FARMLAND

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ABSTRACT. *The distribution of soil preferential paths is a natural process with highly spatiotemporal variability. However, an adequate analysis method to evaluate the spatial structures of soil preferential paths from the viewpoint of ecology has not yet been developed. Hence, the quantification of the complicated distribution patterns of soil preferential paths and their relation to biota activities from the ecological perspective remains elusive. This article presents a new method that combines the image processing technology of morphology and spatial pattern analysis (i.e., quantitative parameters W_b , M_b , U_b , and $g(r)$ function) of landscape ecology to quickly observe and quantify the spatial distribution characteristics of preferential flow paths. The study was conducted on soybean fields in the Three Gorges Reservoir Areas in China. Results demonstrated that the proposed method can describe the spatial connectivity, development degree, and rapid water conductivity of different preferential flow paths (paths of 2.5-5.0 mm were the highest, whereas paths of 1.0-2.5 mm were the lowest). All the preferential flow paths, except for the paths of 1.0-2.5 mm (random distribution pattern), mainly demonstrated a gradual tendency to follow a clumped distribution pattern and community structure relative to soil depth. In addition, all the paths showed low spatial complexity because of their uniform composition relative to soil depth. Soil intra-structure (<1.0 mm, 1.0-2.5 mm, 2.5-5.0 mm, and >5.0 mm) and plant roots effectively influenced the distribution patterns and formation of the preferential paths in the deep farmland soil. Plant roots, in particular, significantly influenced the formation and distribution of the preferential flow paths of 1.0-2.5 mm in the soybean field. Compared with existing methods (e.g., dye tracing, resin impregnation, water breakthrough curve analysis, and computed tomography technique), the proposed method can provide more significant spatial structure information from an ecological perspective. Such information can help expand our understanding of the spatial structures of preferential flow paths and the underlying ecological formation processes in soil ecosystems. The method is also effective, convenient, rapid, and economical when used to study the spatial structures of soil preferential paths at different field scales for soil management.*

Keywords. *Landscape ecology, Morphology, Preferential flow paths, Spatial association, Spatial patterns.*

The term “macropore” has been widely used in the soil literature to describe the soil pores and fissures formed by cracks or soil shrinkage and swelling (Abou Najm et al., 2010), wormholes (Sander and Gerke, 2007), root channels (Ghestem et al., 2011), and finger channels resulting from an unstable wetting front (Helling and Gish, 1991; Hardie et al., 2011). Thus, many different viewpoints about macropore size have emerged in previous studies. Warner et al. (1989) used a computed tomography (CT) scanning technique to obtain a

macropore size larger than 1 mm. Using AUTOCAD and an automatic image analysis technique, Singh et al. (1991) pointed out that macropores are larger than 1.6 mm. However, some researchers also stated that macropore size ranges from 0.1-0.4 mm according to the results of the water breakthrough curve method (Radulovich et al., 1989).

Preferential flow paths are defined as single or interconnected macropores (Abou Najm et al., 2010). Hence, the spatial structure characteristics of soil macropores can represent the spatial structure of preferential flow paths (Cnudde et al., 2006; Abou Najm et al., 2010). Preferential flow paths contribute only a small part to total soil porosity, but they play a key role in many physical processes, such as fluid flow and solute transport (Katuwal et al., 2015). Water and solutes can quickly pass through preferential flow paths without being absorbed by the soil, resulting in significant water loss (Liu et al., 2004) and groundwater pollution (Clothier et al., 2008; Stamper et al., 2014), particularly in farmlands (Sander and Gerke, 2007). To some extent, both water discharge and the process of rainfall (irrigation) recharging groundwater are determined

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by the distribution of soil preferential paths. Moreover, the distribution of soil preferential paths directly influences water and solute transport in porous soil media. Consequently, understanding the ecological formation mechanism of preferential flow paths and determining their distribution characteristics in field and catchment scales are integral to the management of various environmental and agricultural applications.

The spatial structure of preferential flow paths is explored and described by soil scientists through functional soil properties (Velde et al., 1996; Granovsky and McCoy, 1997), soil dye tracer experiments (Flury et al., 1994; Weiler and Flühler, 2004; Tobella et al., 2014), and geometry analysis (situ descriptions or thin sections in the laboratory) (Perret et al., 1999). Among these techniques, the use of digital cameras has been widely practiced to obtain dye images for evaluating the distribution of preferential flow paths on the basis of dye tracer experiments. This method provides direct information on the path number, continuity, depth, and efficiency of different water processes (Trojan and Linden, 1992). For example, Flury et al. (1994) described the transport degree of preferential paths for different farmland treatments through dye images analysis. Janssen and Lennartz (2008) analyzed stained photos to identify the characteristics of preferential flow pathways through paddy bunds. However, previous methods that quantify the spatial structure characteristics of preferential flow paths are all based on model assumptions or image analyses of 2D soil sections. Hence, these methods cannot adequately assess the spatial structure of preferential flow paths. This drawback has prompted the development of a new method that involves 3D image and geometry analysis, that is, the X-ray CT technique (Sander and Gerke 2007; Naveed et al., 2013; Zhang et al., 2014; Katuwal et al., 2015), which can provide in detail the visual, quantitative, non-invasive, and non-destructive geometry of the spatial variation of preferential flow paths. A few studies have employed this method to describe the regions of different path densities (Kastell et al., 2000) and obtain the 3D distribution of large soil pores (Sander et al., 2008). In addition, the CT technique can be adopted to quantify the spatial characteristics of preferential flow paths and correlate these characteristics with hydraulic parameters (Javaux et al., 2006; Naveed et al., 2013). However, this method has several limitations: (1) The specifications of CT scanning facilities and small soil sample size (1-100 cm) (Cnudde et al., 2006; Sander et al., 2008; Zhang et al., 2014) hinder the study of the soil preferential paths of large-scale systems in the field. (2) The formation of soil preferential paths is a natural process (Bronick and Lal, 2005; Abou Najm et al., 2010), and although the CT technique can determine the physical characteristics of preferential flow paths, the inner relations between biota activities and the formation of preferential flow paths are not considered. (3) The CT scanning process is more costly and labor-intensive compared with other methods, and test results cannot be quickly obtained if studies employing this method are conducted in the field.

All the methods mentioned above are based on the physical and chemical analysis perspectives. Nevertheless, the natural process is important in the formation of preferential flow paths, but no study has adopted an ecological approach to describe the spatial structure of preferential flow paths. In the present work, we present a new method that combines the positions of macropores from classical 2D images with the pattern analysis method of landscape ecology to (1) determine the numerical and spatial structure characteristics of different preferential flow paths and to (2) verify whether this new method can be used to analyze the distribution of preferential flow paths.

MATERIALS AND METHODS

FIELD SITE

The field experiment was carried out in farmlands of Simian Mountain (28°23'00" N, 106°37'24" E) with an agricultural history (soybean, 10 years, and conventional tillage) in Jiangjin County, Chongqing Province, Three Gorges Area, China. The annual mean temperature in the Simian Mountain was 18.4°C with an average precipitation of 1,096.7 mm·a⁻¹. The Simian Mountain elevation ranges from 500 to 1,780 m, and the soil textural characteristic is mainly sandy loam (Soil Survey Staff, 2010).

EXPERIMENTAL DESIGN

The soybean plots set previously for a dye tracer experiment in July 2014 were selected for the current work. In the location of these soybean plots, three equidistant plots (named C1, C2, and C3) were set up with the same orientation (northwest), elevation (1,180 m), soil management (traditional tillage artificial method), and slope (3°). For a single sample plot, a 60×60×20 cm steel frame was pushed into the soil to a depth of 20 cm. The soil surface inside the frame was covered with a plastic sheet to preprocess and prevent rainfall from directly eroding the soil surface. The plastic sheet was removed after 48 h, and 9.6 L of brilliant blue FCF (Cheng et al., 2011) dye solution with 4 g·L⁻¹ concentration was slowly sprayed on the soil surface in the frame at a rate of 7.5 mm·h⁻¹ using a spraybar with a cone nozzle (GP1000, Baoding, China) above the center of the frame. A constant flow pump (BT100-02, Baoding, China) was connected to the spraybar to transmit the dye solution. The whole spraying process took approximately 3.6 h. The entire frame was carefully and quickly covered with another plastic sheet to prevent evaporation after the dye solution infiltrated the soil.

The plastic sheet and frame were removed after 24 h. However, the soil edge in the frame was disturbed when the frames were moved. Thus, only the core part of the soil was analyzed. The core region of the dye area (50×50 cm) was excavated horizontally at 10 cm deep intervals until no dye-stained soil layer was visible. Each soil layer was vertically photographed using a digital camera (500D, Canon, Japan) with a 35 mm focal length. An umbrella was used to protect the plot if necessary. The schematic layout of the soil-dyed profiles is shown in figure 1.

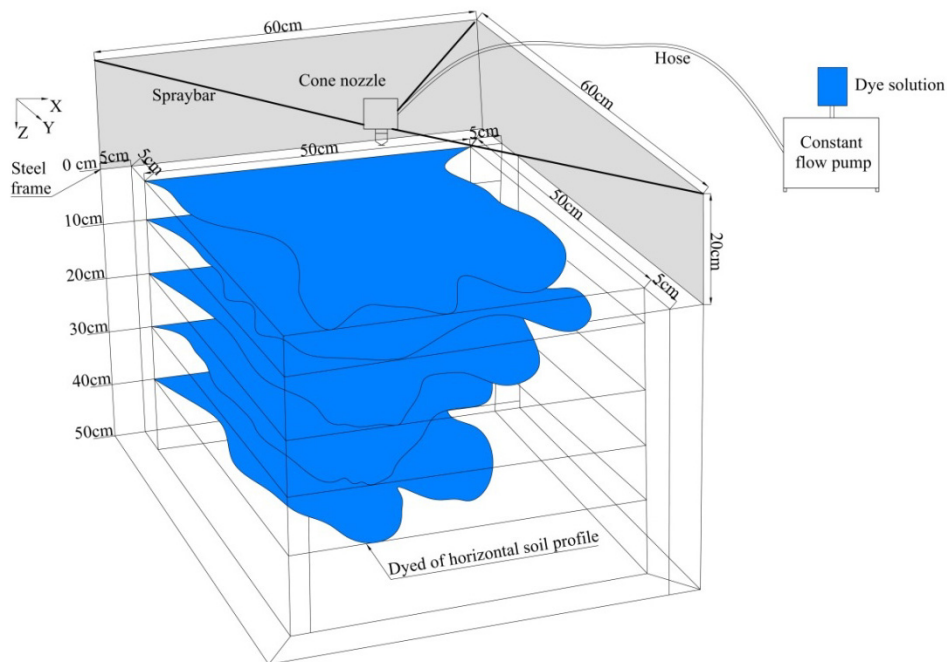


Figure 1. Schematic layout of the dyed soil profiles at one sampling position. Note: In this study, the highest dye depth was 40 cm.

MACROPORES FOR EXTRACTION

Despite the lack of a consensus on their definition, macropores are conceptualized in terms of function and effects (i.e., solute preferential transport was observed in the dye solution, and this condition led to stained soils in this study); such conceptualization is objective, accurate, and comprehensive, as indicated in previous soil science works (Weiler and Flühler, 2004; Abou Najm et al., 2010; Larsbo et al., 2014)

The extraction process in the present work involved four steps (fig. 2). First, each 2D digital image was corrected and tailored with the ERDAS Imagine v9.3 software (Erdas Inc., Atlanta, Ga.) to a size of 50×50 cm, which was equivalent to 500×500 pixels (one pixel corresponded to a square measuring 1×1 mm in the horizontal section). Second, the color images were transformed to binary images by selecting a threshold (i.e., the gray value was 15) and color replacement function. To determine the threshold through the visual discrimination method, the gray value was defined when the binary image almost completely overlapped the stained areas of the original image. Each pixel in the image had only one value (i.e., 0 or 255). Thus, the dyed areas and non-dyed areas could be clearly distinguished from each other. The dyed areas were made up of many 0 pixel values, which were equivalent to the color black. The black areas generally corresponded to macropores, which appeared black in the image. In contrast, the non-dyed areas were made up of 255 pixel values, which were equivalent to the color white. Third, the Image-pro Plus v6.0 software (Media Cybernetics Inc., Rockville, Md.) was utilized to process morphological operations (erosion and dilation functions) and consequently reduce the noise produced by the two previous steps for the accurate calculation of the dyed areas. Fourth, the segmentation threshold and watershed functions

were set to divide the connective dyed areas into independent individuals. Moreover, the size and count functions in the Image-Pro Plus 6.0 software were used to distinguish the diameter (D) four classes (<1.0 mm, 1.0-2.5 mm, 2.5-5.0 mm, >5.0 mm) and the spatial position messages of the macropores in the dyed areas. An artificial statistic method was combined with the ERDAS software to obtain the positions of the plant roots in the corrected 2D digital images.

SPATIAL STRUCTURE ANALYSIS

Natural processes play a significant role in macropore formation, and the structure of macropores characterizes the distribution of preferential flow paths. Thus, we perceive a macropore as a species. In this experimental study, we considered the horizontal stained soil profile as a whole space and the dispersive macropores as individuals of such space. The spatial pattern analysis of landscape ecology can be adopted to conduct systematic research on the spatial structure distribution of preferential flow paths.

Structure characteristics analysis. Many parameters (Getzin et al., 2006; Sterba, 2008; Ruprecht et al., 2010) are widely used to analyze the structural attributes of species at different scales. By contrast, an approach based on the relationships between groups of nearest neighbor species (i.e., uniform angle index, mingling, and dominance parameters) (Hui et al., 2004, 2010; Pastorella and Paletto, 2013; Li et al., 2014) can effectively express whole or unilateral structures and describe two or more aspects of the structural attributes of species.

The uniform angle index (W_i) is a new parameter to describe the spatial distribution pattern of species on the basis of neighborhood relationships (fig. 3). W_i is defined as the proportion of angle α that is smaller than the standard angle α_0 (standard angle is 72°). It is expressed as follows:

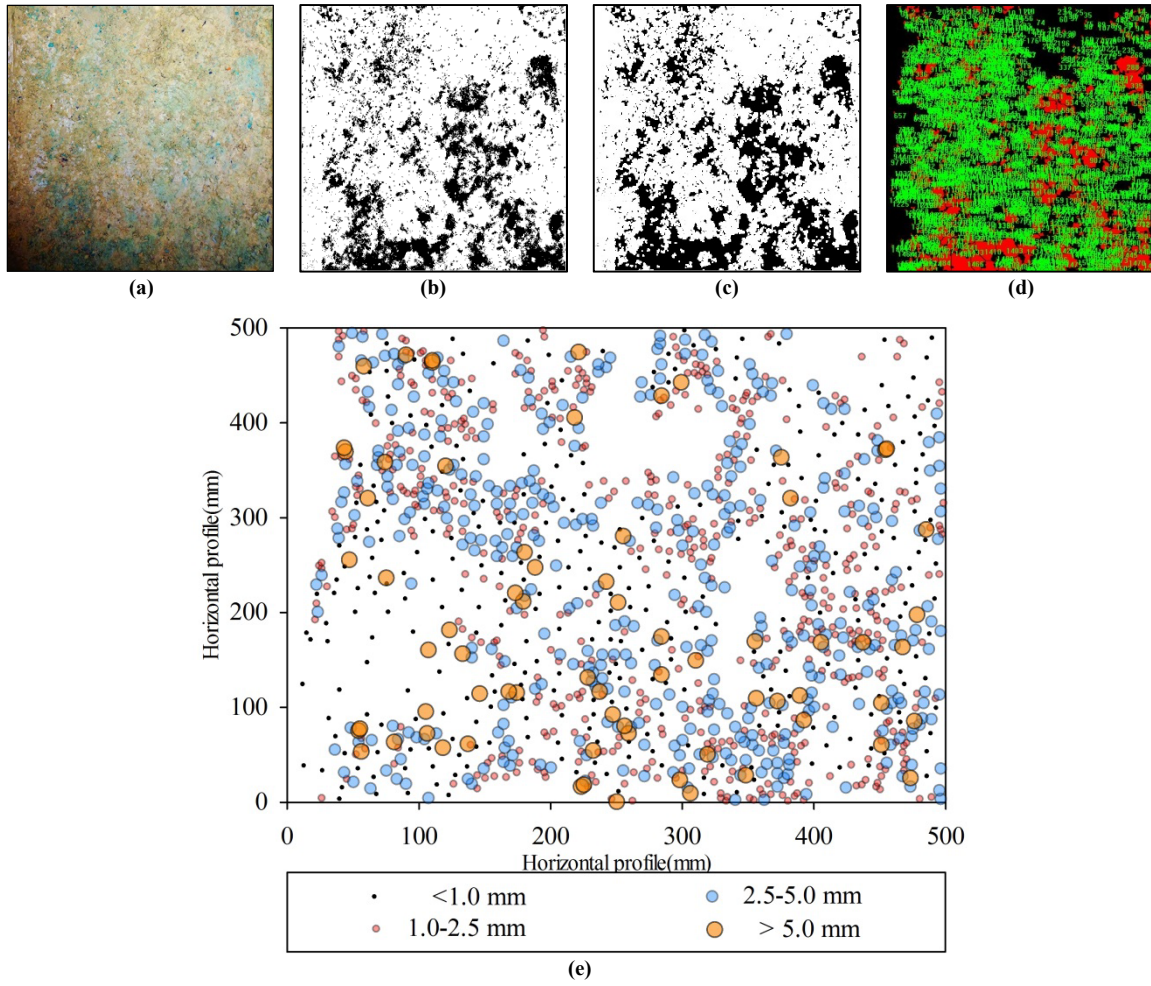


Figure 2. Example of macropores for extraction. (A) Geometrically corrected digital image; (B) binary image (0 and 255); (C) morphological image; (D) image counting process; and (E) positions and quantities of macropores with different size classes.

$$W_i = \frac{1}{n} \sum_{j=1}^n z_{ij} \quad (1)$$

where $z_{ij} = 1$ when angle α is smaller than α_0 and $z_{ij} = 0$ when otherwise $n = 4$ refers to the four nearest neighbor species in this study. Most studies employ a mean value $\bar{W}_i$ to identify the overall distribution pattern of a species. $\bar{W}_i$ values fall in the interval 0.475-0.517, and the distribution pattern tends to be random. When $\bar{W}_i > 0.517$, the distribution pattern tends to clump, whereas when $\bar{W}_i$ values < 0.475 , the distribution pattern tends to be regular.

Mingling (M_i) is used to express the spatial configuration pattern of different species in a group of multi-species (i.e., the degree of mixture). M_i is also defined as the proportion of the n nearest neighbor species that differ from the reference species. It is expressed as follows:

$$M_i = \frac{1}{n} \sum_{j=1}^n v_{ij} \quad (2)$$

where $v_{ij} = 1$ when neighbor species j is not that same as the reference species i and $v_{ij} = 0$ when otherwise. Here, $n = 4$ refers to the four nearest neighbors in this study. M_i ranges from 0 to 1. Values that are close to 0 indicate that the nearest neighbor species and reference species belong to the same species in the group. This characteristic suggests a low level of species mingling. By contrast,

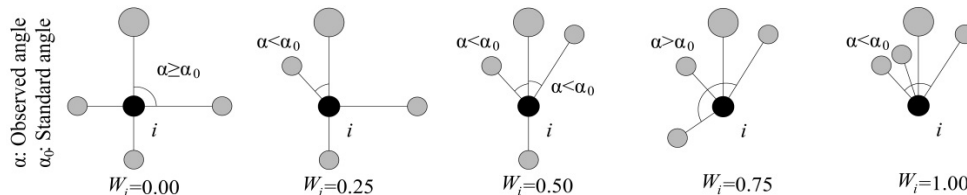


Figure 3. Five forms of the uniform angle index. Left to right: very regular, regular, random, clumped, and very clumped distribution patterns.

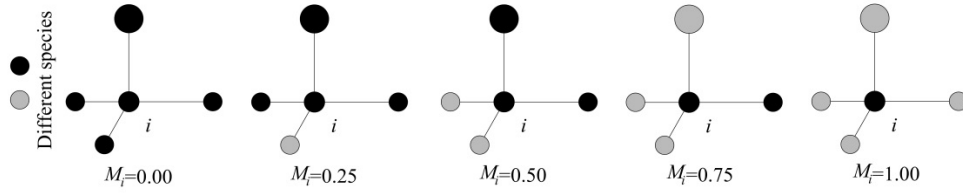


Figure 4. Five forms of mingling. Species composition or spatial configuration patterns (L to R): no mixture, low mixture, medium mixture, high mixture, and complete mixture.

values close to 1 indicate that the reference species occur in isolation. This condition implies a high level of species mingling (fig. 4).

In contrast to the concepts of uniform angle index (W_i) and mingling (M_i), dominance (U_i) (i.e., predominance of species) means that the proportion of the n nearest neighbor species of a given reference species is smaller than the reference species. Dominance is expressed as:

$$U_i = \frac{1}{n} \sum_{j=1}^n k_{ij} \quad (3)$$

where $k_{ij} = 0$ when neighbor species j is smaller than the reference species i and $k_{ij} = 1$ when otherwise. Here, $n = 4$ refers to the four nearest neighbor species in this study. The U_i value reflects the biophysical position of different species and can assume five probable values when the number of nearest species is four (0.00, 0.25, 0.50, 0.75, and 1.00). A U_i value that is close to 0 and 1 indicates that

the species is predominant and absolutely disadvantaged, respectively (fig. 5).

Spatial structural analysis was carried out using the Crancod software (Crancod v1.4).

Spatial relationship analysis. The pair correlation $g(r)$ function (Diggle, 2003), which was derived from Ripley's K function (Ripley, 1977), is the probability of observing a pair of points separated by distance r divided by the corresponding probability for a Poisson process. It is expressed as:

$$g(r) = \frac{1}{2\pi r} \cdot \frac{dK(r)}{dr} \quad (4)$$

where r is the radius of the circle (mm) and d_r is the width of the circle (mm). Most studies employ the positional relation between the $g(r)$ function and the envelope lines (fig. 6). This step was performed in this study using the R program (R v3.2.0).

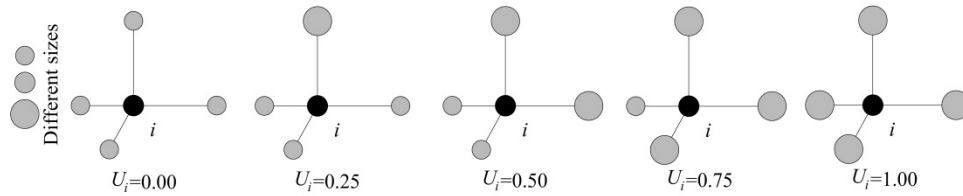


Figure 5. Five forms of dominance (L to R): predominant, subdominant, medium, disadvantaged, and absolutely disadvantaged.

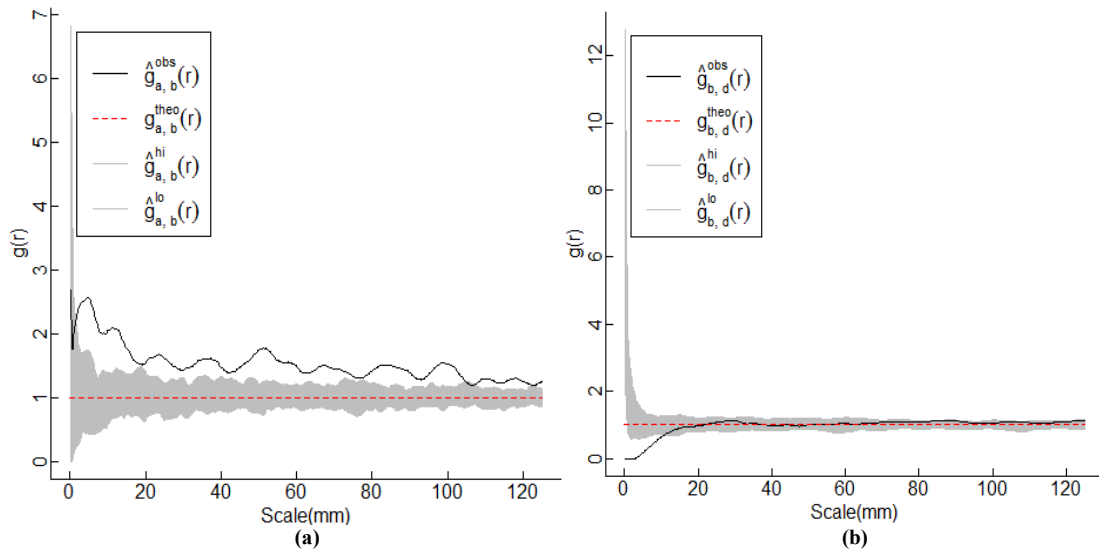


Figure 6. Example of spatial correlation analysis. The relationship is positive when the $g(r)$ function is above the high envelope line (A graph). The relationship is negative when the $g(r)$ function is below the low envelope line (0-20 mm in the B graph). A $g(r)$ function between the high envelope line and the low envelope line indicates no relationship (20-125 mm in the B graph).

STATISTICAL ANALYSIS

MS Excel 2010 (Microsoft Corp., Redmond, Wash.) was utilized for the data calculation. A one-way analysis of variance (ANOVA) was performed with the SPSS v18.0 software (SPSS Inc., Chicago, Ill.) at a 0.05 level of significance to determine whether the number of macropores varied with soil depth (table 2).

RESULTS AND DISCUSSION

NUMERICAL CHARACTERISTICS OF MACROPORES

The top soil (0 cm) was entirely stained with a brilliant blue dye because water is in a non-equilibrium state in the flow field (Flury et al., 1994). That is, macropores could not be extracted from the surface soil in the plots. In this study, the deepest stain reached 40 cm. The results were detected in four horizontal soil profiles (10-40 cm) in each plot, and 12 soil profiles were obtained in the three soybean plots.

The numbers of macropores at different depths are listed in table 1. The variance analysis indicated significant differences between the soil layers, as shown in table 2 ($P < 0.05$). This result demonstrated the significant change and decrease in the number of macropores along with soil depth in the plot. The trends observed in the three plots were similar, i.e., the number of macropores quickly decreased at the top 10-20 cm and decreased gradually up to a depth of 40 cm, and the number of surface soil macropores was larger than that at the bottom (table 1). The total numbers of macropores in the three plots were 2.38, 2.13, and 2.31 times greater at the depths of 0-20 cm than at the depths of 20-40 cm. This result is consistent with the findings of Luo et al. (2008), who used CT scanning to analyze the overall macroporosity in the soil column (diameter of 10 cm and vertical length of 30 cm) of a farmland, which decreased dramatically from approximately 15% at the surface to roughly 2% at a depth of 30 cm. Zhang et al. (2014) also determined a few macropores in a plow pan (soil layers 20-35 cm deep) with the CT scanning technique.

For all the plots, the most common macropore diameters were from 1.0-5.0 mm (fig. 7). The macropores of 1.0-2.5 mm dominated all plots, followed by the macropores of 2.5-5.0 mm and <1.0 mm. These results may be attributed to the diameters of the plant roots mostly measuring 1.0-

Table 2. Variation in the number of macropores in the soil layers relative to soil depth (mean $\pm$ SE).

Soil Depth (cm)	Diameter (mm) ^[a]				Total No. of Macropores
	<1.0	1.0-2.5	2.5-5.0	>5.0	
10	317 $\pm$ 20.2 ^a	545 $\pm$ 31.8 ^a	371 $\pm$ 28.0 ^a	78 $\pm$ 4.6 ^a	1311 $\pm$ 22.7 ^a
20	213 $\pm$ 16.5 ^b	407 $\pm$ 53.1 ^b	273 $\pm$ 23.1 ^b	48 $\pm$ 3.0 ^b	941 $\pm$ 74.5 ^b
30	154 $\pm$ 20.3 ^c	202 $\pm$ 27.0 ^c	214 $\pm$ 15.3 ^b	25 $\pm$ 2.9 ^c	594 $\pm$ 19.1 ^c
40	115 $\pm$ 10.2 ^c	136 $\pm$ 26.3 ^d	133 $\pm$ 9.9 ^c	12 $\pm$ 2.4 ^d	397 $\pm$ 19.8 ^d

[a] Different lowercase letters in the same column indicate a significant difference among the soil layers. ($P < 0.05$, $n=3$)

2.5 mm in the three plots. These outcomes are consistent with the previous findings of Iversen et al. (2012), who used image analysis method to report that the most common macropore diameters were 2.0 and 3.0 mm in farmland.

On the basis of the results (table 1), we determined that the macropores with a diameter ≥ 1.0 mm accounted for 73.1% to 78.2% of the total macropores. Such finding is similar to that of Zhang et al. (2014), who analyzed the 3D images of a scanned soil column (10 cm in diameter and 20 cm in height) from agricultural plots and found that the macropores with a diameter > 1.0 mm accounted for 61.0% to 81.0% of the total macropores.

SPATIAL DISTRIBUTION OF MACROPORES

The macropores quantified in all the plots were displayed for spatial visualization. However, selected example C1 plot is shown in figure 8 (C1, C2, and C3 have the same spatial pattern). The macropores in the C1 plot were mostly concentrated in the upper 20 cm of the soil, particularly at the 0-10 cm soil depth (fig. 8). This result is similar to that of a previous study that adopted the CT technique and found that the large macroporosity at the top 5 cm corresponded to the harrowed layer (Naveed et al., 2013; Larsbo et al., 2014). As shown in the density images of the C1 plot (fig. 8), the spatial distribution of the macropores of different sizes distinctly varied. For example, the image of the macropore with a diameter > 5.0 mm exhibited aggregative conditions with six dark areas (density value of 4-5) at a depth of 10-40 cm. The dark areas (i.e., color from blue to red) indicated a large density value (varying from 1 to 5), which denoted a clumped macropore layout. The figure also showed that only part of the macropores on the subsurface continuously appeared from the top soil to the bottom soil. This characteristic was indicative of the ineffectiveness of some of the macropores in flow and transport. Similarly, Luxmoore et al. (1990) and Luo et al. (2008) showed that not all macropores are hydrologically active in forest watersheds and farmlands. The figure 8D revealed that six primary macropore areas penetrated through the 10-40 cm soil layers. The spatial visualizations and density images of the different macropore distributions indicated that the connectivity of macropore networks considerably varies with soil depth. In the present work, we compared the macropores with diameters of <1.0 mm, 1.0-2.5 mm, and > 5.0 mm. We observed that the networks of macropores with diameter of 2.5-5.0 mm tended to be more vertically oriented and continuous compared with the other networks of macropores of different sizes. Nevertheless, the macropores of 1.0-2.5 mm exhibited the worst

Table 1. Quantitative characteristics of macropores relative to soil depth.

Plot	Soil Depth (cm)	Diameter (mm)				Total No. of Macropores
		<1.0	1.0-2.5	2.5-5.0	>5.0	
C1	10	319	555	411	70	1355
	20	210	512	319	49	1090
	30	121	238	212	22	593
	40	108	189	128	11	436
C2	10	281	595	317	86	1279
	20	243	339	246	42	870
	30	191	218	188	31	628
	40	135	110	119	17	381
C3	10	351	486	385	78	1300
	20	186	371	254	52	863
	30	149	149	241	23	562
	40	102	110	152	9	373

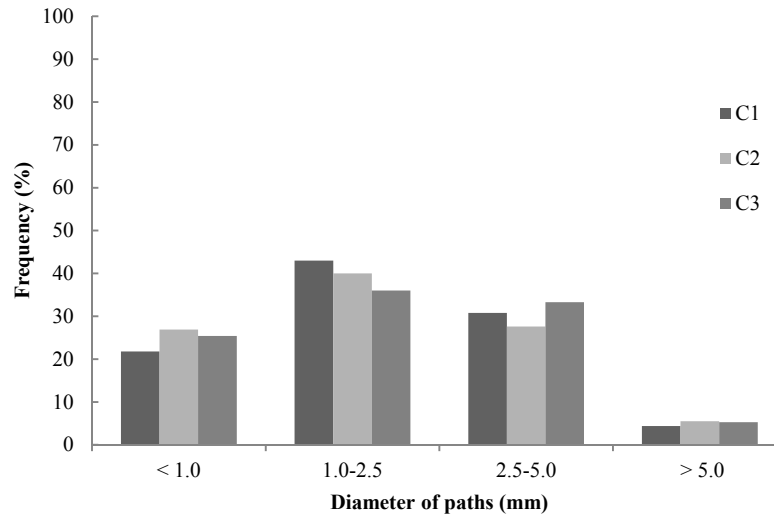


Figure 7. Frequency of different macropore sizes (<1.0 mm, 1.0-2.5 mm, 2.5-5.0 mm, and >5.0 mm).

condition. As reported in previous studies (Jarvis, 2007; Luo et al., 2010; Palmer et al., 2011; Larsbo et al., 2014), preferential transport is expected to occur in soils containing large continuous macropores, and the connectivity of different macropores indicates the development degree of preferential flow paths and the predominant water transfer process. Our results indirectly indicate that the path which consisted of macropores of 2.5-5.0 mm exhibited a highly preferential transport capability and hydraulic conductivity.

The spatial distribution of macropores affects their ability to transport water and, consequently, the nutrient concentrations in runoff (Pachepsky et al., 2000), particularly in farmlands (Luo et al., 2008). In the present study, the connectivity of preferential flow paths was also characterized by quantifying both macropore interconnectivity and macropore path through the entire soil. The connectivity and development degrees of the preferential flow paths of different sizes were arranged as follows from high to low: 2.5-5.0 mm, <1.0 mm, >5.0 mm, and 1.0-2.5 mm (C1, C2, and C3 showed the same results). It can be deduced that the water infiltration rate was significantly higher in the preferential flow paths of 2.5-5.0 mm than in the other preferential flow paths.

SPATIAL STRUCTURE CHARACTERISTICS OF PREFERENTIAL FLOW PATHS

The distribution patterns of preferential flow paths have been widely studied in the context of macropore structure, shape, and pattern (Abou Najm et al., 2010; Luo et al., 2010). Thus, the distribution of macropore spatial structure reflects the distribution pattern of preferential flow paths.

In the three experimental plots in this work, the $\bar{W}_i$ values of the macropores of different sizes gradually increased with soil depth (table 3), resulting in macropores gradually exhibiting a clumped ($\bar{W}_i$ values larger than 0.517) distribution pattern, except for the macropores measuring 1.0-2.5 mm. The $\bar{W}_i$ values of the macropores measuring 1.0-2.5 mm fell in the interval 0.475-0.517; these macropores were randomly distributed at depths of 10-40 cm in all three plots. However, previous studies (fig. 8) showed that distribution patterns range from the dispersing type to the centralized type and easily form the upper and lower connectivity of paths. This observation could be the reason why the macropores of 1.0-2.5 mm in the present work exhibited the worst continuous condition (fig. 8). Although the macropores of 1.0-2.5 mm exhibited a random distribution pattern, their $\bar{W}_i$ values gradually approached the 0.517 tendency with soil depth. Moreover, the $\bar{W}_i$ values of all the macropores in each layer were 0.488-0.568, 0.496-0.574, and 0.492-0.582 for the C1, C2, and C3 plots, respectively. To some extent, the entire macropore spatial structure tended to feature a gradually clumped (from random to clumped pattern) distribution pattern from the surface to the deep layer. This result reflected the spatial distribution of preferential paths was clumped at the deeper layer. This observation may help explain how the tillage management destroyed the topsoil structure (0-10 cm). By contrast, a continuous and interconnected intra-aggregate pore network was observed for the no-till aggregates, and this characteristic indicated a

Table 3. Averages of the mean uniform angle indexes for different macropores in each soil layers (10, 20, 30, and 40 cm) of the soybean fields (C1, C2, C3).

Plot	Soil Depth (cm)	Diameter (mm)				All
		<1.0	1.0-2.5	2.5-5.0	>5.0	
C1	10	0.520	0.477	0.521	0.479	0.488
	20	0.536	0.490	0.524	0.544	0.520
	30	0.550	0.492	0.543	0.546	0.559
	40	0.594	0.510	0.553	0.542	0.568
C2	10	0.533	0.479	0.537	0.527	0.493
	20	0.542	0.483	0.555	0.547	0.517
	30	0.561	0.496	0.556	0.561	0.546
	40	0.613	0.512	0.580	0.645	0.574
C3	10	0.541	0.491	0.544	0.541	0.482
	20	0.541	0.501	0.547	0.580	0.519
	30	0.564	0.508	0.560	0.585	0.548
	40	0.570	0.511	0.592	0.775	0.582

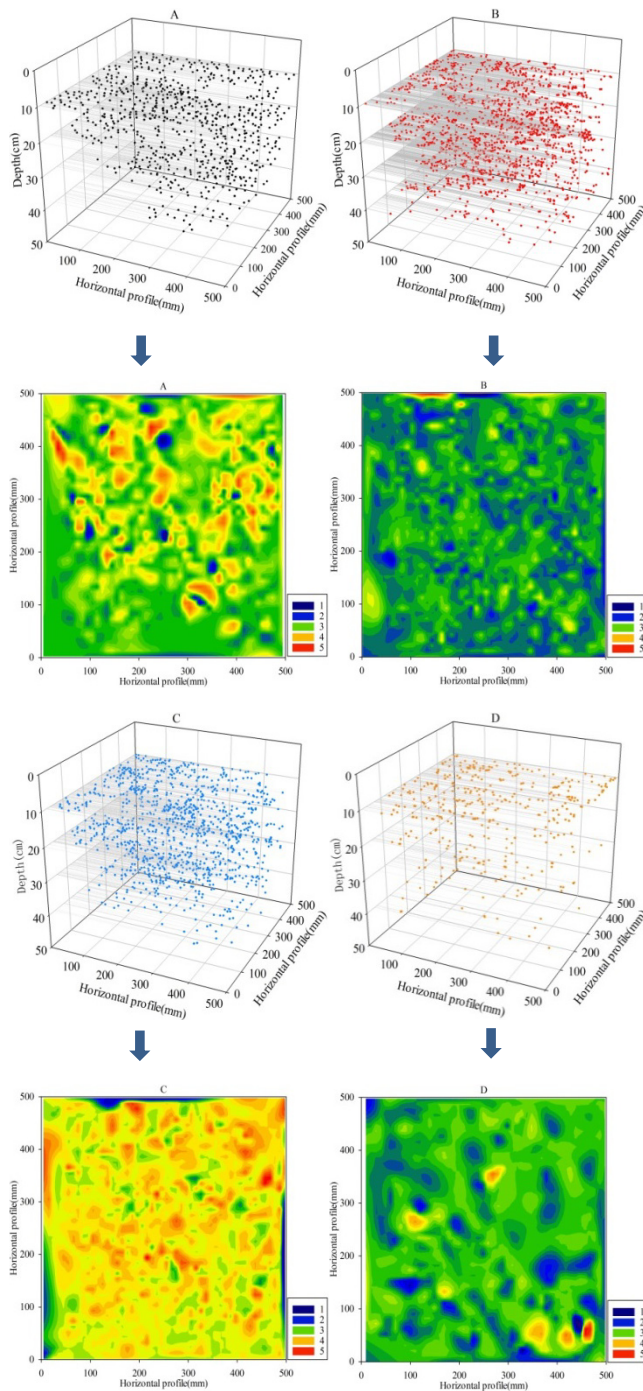


Figure 8. Examples of C1 plot, with distribution status and density images of macropores of different sizes (A: <1.0 mm, B: 1.0-2.5 mm, C: 2.5-5.0 mm, D: >5.0 mm) in the 10-40 cm soil layer.

well-structured aggregate pattern, as reported in the literature (Smucker et al., 2007). Furthermore, Kung et al. (2005) used the measured tracer breakthrough curve method to obtain the preferential pathways that were naturally clustered into domains of a farmland.

On the basis of a solute breakthrough curve analysis, Larsbo et al. (2014) analyzed the relationship between the macropore network characteristics and the degree of the preferential solute transport of a farmland and determined that the macropores were homogeneously distributed and

numerous, which were indicative of a low degree of preferential transport. Combined with the results of a uniform angle index analysis, this result implies that preferential flow mainly occurs beneath the topsoil of farmlands. This outcome is consistent with that of previous studies (Chen et al., 2015).

Our study also supports the findings of Sander et al. (2008), who used CT to obtain the 3D distribution of large pore systems and found aggregated macropore structures in the subsoil of their samples. Similar results were observed by Larsbo et al. (2014), who adopted a global connectivity measure with 3D image analysis and found macropore clusters connected from the soil surface to the bottom. Some factors that may influence macropore distribution patterns include roots, earthworms, soil compaction level, field management, and type of agriculture machinery used. Abou Najm et al. (2010) utilized liquid latex to visualize the soil preferential paths in a field and found that preferential path structures are more complex in the soil surface than in the deep layer because in their work, the tillage damage to the inter-aggregate structure of the soil in the plow pan resulted in paths that clumped with depth. In our study, we also observed an aggregate distribution pattern in the plots relative to depth. The mean uniform angle index ($\overline{W}_i$) can directly reflect the distribution patterns of macropores from the surface to the deep soil layer.

However, the uniform angle index only shows the overall distribution pattern of macropores relative to soil depth; that is, it does not like as the mingling parameter to illustrate the detailed distribution patterns and spatial complexity of macropores from different size classes. One feature of the distribution frequency common to all the three plots was that almost all the macropores were part of aggregations with medium-complete degrees of mixture ($M = 0.50-1.00$). Moreover, more than half (approximately 52.75% to 56.00%) of the frequency values (total values of 2.11, 2.13, and 2.24) of the C1, C2, and C3 plots belonged to the high-degree axis of mingling ($M = 0.75-1.00$). The frequency values of mingling in each soil layer initially increased and declined, accompanied by an increase in mingling for grades ranging from 0.00 to 1.00. Compared with the degrees of mixture ($M = 0.00-0.50$ and 1.00), the frequency values of the degree of mixture ($M = 0.75$) were the largest in each plot, with the maximum value (0.40) observed in the structural composition ($M = 0.75$) of the C3 plot (fig. 9). Evidently, the different macropores exhibited a high mixture pattern or complexity from the surface to the bottom layers of all the experimental plots. Nevertheless, the degrees of mixture that represented the structural complexities of the macropores gradually decreased from the surface to the deep soil layers, as shown in figure 9. On the one hand, the complexity of the macropores substantially decreased because of the low number of macropores (table 1) in the deep soil, which significantly affected the branched and unbranched macropores (Katuwal et al., 2015). On the other hand, tillage management considerably increased the complexity of the topsoil structure, which reduced the path formation. By

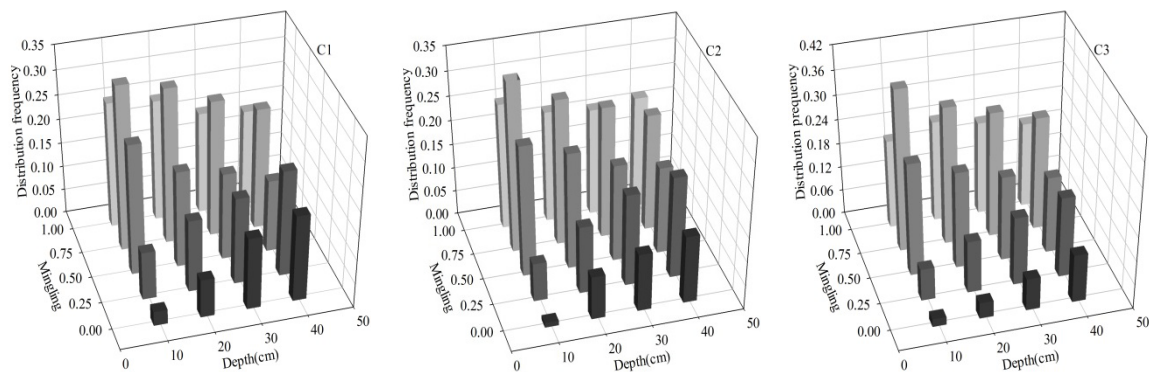


Figure 9. Distribution frequency of the mingling parameter in different soil layers (10, 20, 30, and 40 cm) in each plot (C1, C2, and C3).

comparing conventionally tilled farmlands and grasslands, a study pointed out that soil solutions are more effectively transported to the interior of aggregates by the continuous, less tortuous, and wide flow paths of grassland aggregates (Peth et al., 2008). Using visual liquid latex (i.e., resin impregnation), Abou Najm et al. (2010) also demonstrated that preferential paths are characterized by a complex structure in the field, with such structure becoming more increasingly complex in the soil surface than in the deep layer.

Mingling is further considered as an important parameter to characterize the spatial structure of macropores, which reflects their spatial complexity and distribution pattern. Degrees of mixture close to 0 indicate that macropores have the same size as those in the reference paths occurring in a group. This observation indicates a low level of macropore mingling and a high degree of concentration (fig. 4). As seen in figure 9, the frequency values of the degrees of mixture ($M = 0.00$ and 0.25) in each column apparently increased with depth in each plot and reached the maximum at a mingling value = 0.25 . Thus, the macropores were more clumped in the deep soil layers than in the upper layers. The macropores also demonstrated a gradual formation of a clumped distribution pattern, which is consistent with the pattern observed in our previous study (table 3). Feng and Hao (2000) adopted the CT technique to describe the dispersion degrees of macropores and the differences in macropore sizes in farmland cross sections, all of which gradually decreased with depth. The clumped patterns shown in table 3 can be effectively explained by the mingling parameter analysis.

Generally, a small M indicates that the degree of mixture of soil macropores is low and that the macropore spatial structure in soil is homogeneous. Homogeneous and clumped macropores are advantageous in constructing preferential paths and can easily be transported to deep soil layers. As described previously, the mingling parameter indirectly indicates that preferential paths with a high degree of or improved connectivity occur mainly in the deep soil. Previous studies have also shown through dye tracer experiments that preferential flow, in contrast to uniform flow, makes up a small volume of a soil matrix. Thus, a high degree of preferential flow corresponds to a low total dye coverage (Tobella et al., 2014). Using a dye tracer, Sander and Gerke (2007) found that the plow pan of

an experimental paddy field in southeast China allowed preferential flow toward the shallow groundwater. Chen et al. (2015) also used a dye tracer and revealed that the dyed areas indicating preferential flow (below 80%, van Schaik, 2009) mainly occurred in the subsoil zones (beneath 10 cm) of a southwest China farmland.

In our study, M not only indicated the spatial complexity and distribution of the different paths but also revealed the continuous degrees of the preferential paths relative to soil depth.

To further understand the development degrees of the different preferential paths comprising macropores in soil space, the values of dominance (U) were analyzed in this work. As indicated in figure 10, the U values of the different macropores in the three plots all fluctuated in the vertical direction and changed mostly by 1.0-2.5 mm. This outcome is attributed to the high number of macropores of 1.0-2.5 mm (table 1) relative to the macropores of different sizes. This condition influenced the connectivity of the macropores. Figure 10 shows that the values of dominance for the macropores of 1.0-2.5 mm were approximately 1.35, 1.41, and 1.35 times higher than those for the macropores of 2.5-5.0 mm in the C1, C2, and C3 plots, respectively. Nevertheless, the high values of dominance were close to 1, which indicates a low degree of dominance (Hui et al., 2010; Li et al., 2014) for a species in space. Thus, U can correctly express the development degree for a particular preferential path, which consists of corresponding macropores in soil space, and reflect the composition of macropores in the soil layer. As shown in figure 10, the values of dominance evidently increased relative to soil depth given the same macropore size. This result demonstrates that the composition diversity of macropores decreases relative to soil depth, with diversity becoming relative simple at deep soil layers (i.e., all macropores are of uniform size). These results are consistent with the previous findings of Feng et al. (2002) who used CT and found that macropore structure and composition tend to simplify with the depth of agricultural soils. On the basis of the resulting dominance values that changed with depth, which were similar to the findings of Luo et al. (2010), we determined that homogenous soil macropores were formed below the plow pan of the farmland. The dominance values among the macropores shown in figure 10 were distinctly different for the three plots and could be ranked from high

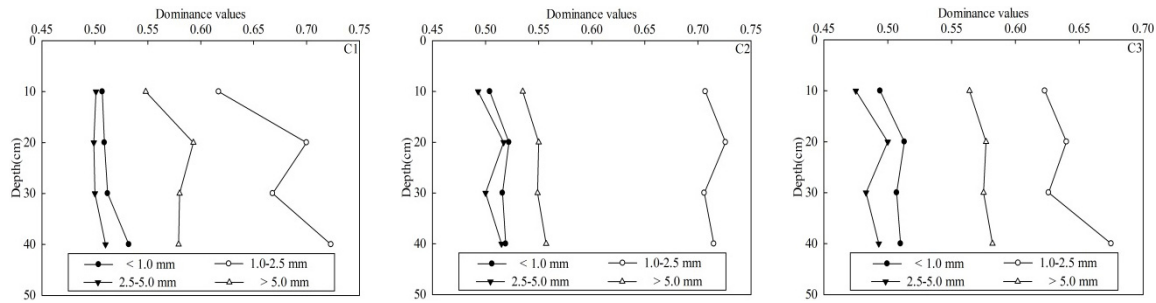


Figure 10. Dominance values in different soil layers (10, 20, 30, and 40 cm) in each plot (C1, C2, and C3).

to low as follows: 1.0-2.5 mm, >5.0 mm, <1.0 mm, and 2.5-5.0 mm. The result also demonstrated the opposite development degree of the preferential paths, which consisted of macropores from different size classes. This result is highly consistent with the previous findings on development degrees (fig. 8), in which the preferential flow paths 2.5-5.0 mm and 1.0-2.5 mm were the highest and lowest, respectively. Katuwal et al. (2015) obtained similar results by CT scanning technique, that is, the size distribution of the macropores in a sandy loam farmland was larger than 2.2 mm. The degrees of dominance that decreased with depth, which can explain the decrease in the development degrees of the preferential flow found in previous studies using dye tracers (Sander and Gerke, 2007; Tobella et al., 2014).

SPATIAL CORRELATION CHARACTERISTICS OF PREFERENTIAL PATHS

The three soybean plots exhibited preferential flow paths with similar spatial correlation characteristics (table 4). Different paths showed no relationship, except for the paths of 1.0-2.5 mm and 2.5-5.0 mm in the 10-20 cm soil layers (table 3). In addition, all the macropores in the 20-40 cm soil layers showed a positive relationship, which indicated that the paths of 1.0-2.5 mm significantly influenced the distribution of the paths of 2.5-5.0 mm. Moreover, the paths of 1.0 mm-2.5 mm played an important role in the formation of the paths of 2.5-5.0 mm. One reason for this observation is that the paths of 1.0-2.5 mm showed the most number of macropores (table 1). Another reason is that similar and small paths can promote the formation of

large preferential paths (Zhang et al., 2014). Similarly, Peth et al. (2008) found that high boundary-connected pore volumes between inter-aggregate and intra-aggregate pores easily formed paths that enhanced the exchange of water and gases. Our results thus reveal the interaction and internal connection among preferential flow paths (Kung et al., 2005). Our data (table 4) also verify that different paths have a positive effect on deep soil layers.

SPATIAL CORRELATION CHARACTERISTICS BETWEEN PREFERENTIAL PATHS AND ROOT ACTIVITIES

The point pattern analysis identifies the spatial distribution characteristics of different species and determines their spatial association (Diggle, 2003). In our study, the roots were observed to reach a maximum depth of 20 cm. Thus, we analyzed the position data of the roots from 0-20 cm (table 5). The roots were positively correlated with the preferential paths of 1.0-2.5 mm in the 10 cm soil layer of the C1 and C2 plots, as well as the paths 2.5-5.0 mm in the 20 cm soil layer of the C3 plot. This characteristic could be attributed to the large amount of roots with a diameter ranging from 1.0-2.0 mm in each plot. Luo et al. (2010) reported that preferential flow paths are formed by highly continuous and round-shaped roots. Therefore, root development and subsequent decay form the corresponding sizes of paths, and this effect is significant in deep soil layers (20 cm soil layer in the C1, C2, and C3 plots). We also observed a positive correlation between the paths and roots in the deep soil layer (table 4). Similar to the outcomes of previous studies, the biopores (root channels) in the present work could significantly and continuously penetrate the soil (Sander et al., 2008).

Biotic and abiotic processes mainly form preferential flow paths. In the present work, the preferential flow paths were highly associated with the root activities, as shown in table 5. This relationship indicates that preferential flow

Table 4. Spatial correlations of different paths (A: <1.0 mm, B: 1.0-2.5 mm, C: 2.5-5.0 mm, D: >5.0 mm) in three plots.^[a]

Plot	Soil Depth (cm)	A and B	A and C	A and D	B and C	B and D	C and D
C1	10	0	0	0	+	0	0
	20	0	0	0	+	0	0
	30	+	+	+	+	+	+
	40	+	+	+	+	+	+
C2	10	0	0	0	+	0	0
	20	+	0	0	+	0	0
	30	+	+	+	+	+	+
	40	+	+	+	+	+	+
C3	10	0	0	0	+	0	0
	20	0	0	0	0	0	0
	30	+	0	+	+	+	+
	40	+	+	+	+	+	+

^[a] Signs "0" and "+" denote no relationship and positive interrelated relationship, respectively.

Table 5. Spatial correlation among preferential paths (A: D <1.0 mm, B: 1.0-2.5 mm, C: 2.5-5.0 mm, D: >5.0 mm) and roots (E: position of roots) in three plots.^[a]

Plot	Soil Depth (cm)	A and E	B and E	C and E	D and E
C1	10	0	+	0	0
	20	+	+	+	+
C2	10	0	+	0	0
	20	+	+	+	+
C3	10	0	+	0	0
	20	+	+	+	+

^[a] Signs "0" and "+" denote no relationship and positive interrelated relationship, respectively.

occurs in the channels of dead or decaying roots and forms around live roots in the cover of vegetation lands (Ghestem et al., 2011). Furthermore, tillage practices are the main factors that affect the preferential flow paths in the 0-10 cm layer. In our work, the preferential paths most probably evolved because of the combined effects of tillage, biodegradation of the roots in the subsoil layers, and tillage damage to the inter-aggregate structure of the soil (Abou Najm et al., 2010). By contrast, the roots correlated positively with the different preferential paths at the deep soil layer. Researchers have found that densely distributed root systems are beneficial in the formation of preferential flow paths in soil and in the vertical transport of soil moisture (Gish et al., 1998; Tobella et al., 2014). Considering these characteristics, we can deduce that plant roots positively affect water and nutrient transport in farmlands.

CONCLUSION

The approach proposed in this study allows a detailed evaluation and characterization of the diverse morphological features of preferential flow paths consisting of macropores through landscape pattern analysis. A suite of spatial pattern analysis parameters (i.e., W , M , U , and $g(r)$ function) was quantified for 12 soil-dyed images from the ecological viewpoint. Both the W and M parameters describe the soil porosity system and indicate the distribution patterns and composition of paths consisting of macropores. In this work, the structural distribution of the preferential flow paths showed a gradually clumped pattern, and the complexities decreased with soil depth. Hence, the concentrated distribution pattern and low mixture degree (large number of preferential flow paths of the same size) easily resulted in highly continuous and effective preferential flow paths and increased the water infiltration rate of the paths. The U parameter combined with the density images indicated the highest spatial connectivity and development degree of the preferential flow paths of 2.5-5.0 mm. The $g(r)$ function determined some quantitative relationships between the distribution of the preferential flow paths and the plant roots. In particular, the root significantly influenced the size of the preferential paths (1.0-2.5 mm).

The proposed method is easier to implement in the field compared with existing and established techniques (e.g., dye tracing, resin impregnation, water breakthrough curve analysis, and CT technique). The proposed method also accurately evaluates the spatial structure characteristics and development degrees of preferential flow paths. From the management perspective, this method is more economical and convenient than other techniques for analyzing soil structure. The benefits of the method also compensate for the deficiency of the current technology in the theoretical analysis of preferential flow path structures. Applying the proposed method further clarifies the reasons for the formation and distribution of the spatial structure of soil preferential flow paths from an ecological perspective. However, this method has serious limitations. (1) This

method cannot be used to analyze small soil samples (e.g., soil column) at the laboratory scale because of the limited image resolution in this study. (2) Unlike the CT scanning technique, the proposed method cannot visualize 3D spatial distribution images. Nevertheless, landscape pattern analysis will supplement and advance existing research from the ecological perspective, which has yet to replace existing technology.

In future studies, we will combine a non-destructive x-ray CT technique and 3D image analysis with the proposed method (i.e., landscape pattern analysis) for the comparative analysis of the spatial structure of preferential paths in an undisturbed soil column sample for a broad range of land use types (e.g., grassland, forestry, orchard, wasteland), soil types, and management practices. Quantifying these relationships is necessary to enhance our ability to predict the changes in soil ecosystems caused by management and global changes.

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